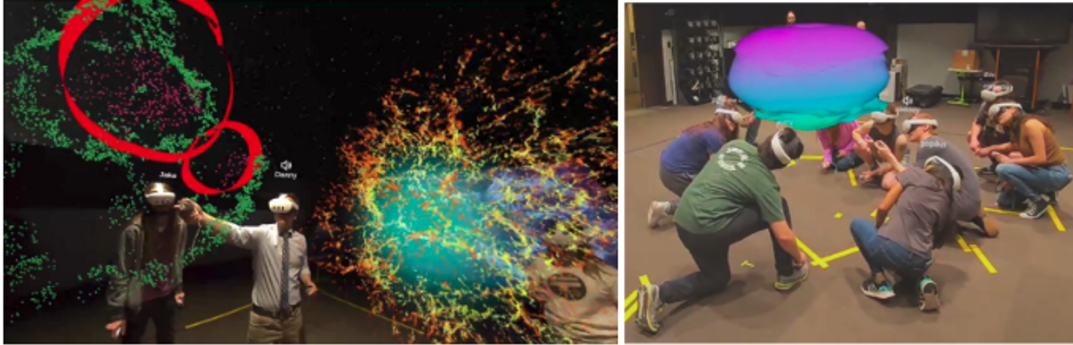


# CollabXR: an open-source platform for immersive spatial communication of learning materials

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**Figure 1: CollabXR classroom use in Physics and Astronomy at Purdue University observing reconstructions of supernova remnants (left) and simulations of galaxy plasma bubbles (right)**

## Abstract

CollabXR [1] is a multi-user visualization and interaction platform built for shared instruction, training and research through immersive engagement with three-dimensional materials. Developed at the Envision Center - Purdue University, CollabXR was originally designed for VR supernova remnant visualizations for astrophysics [2]. In the last 2 years, the CollabXR platform has expanded into co-located MR for any domain that benefits from shared observation, instruction and exploration 3D data. Through the last seven years of leveraging consumer headsets and adapting to remote shared VR lectures during Covid, Purdue University has been building up the capacity to provide immersive lectures in a scalable shared environment. The motivation of building this platform stemmed from the desire to explore complex scientific data in a collaborative platform, something that has been difficult to accomplish with traditional scientific visualization software or general-purpose MR applications. Added to this is the communication challenge which arises when discussing or sharing findings from the explorations of the materials in 2D or asynchronous VR experiences. This “visualization and communication gap” led us to build a system to enable natural perceptual processes and embodied interactions with material where spatial communication of virtual content is intuitive and optimized for classroom use.

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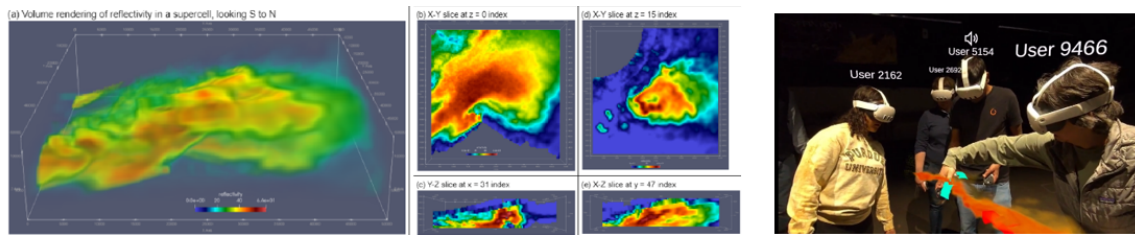
## 1 Background

CollabXR is an experimental, co-located, multi-user platform that allows students and instructors to engage with spatial data collaboratively in a shared physical environment. Currently compatible with both Meta Quest 3 and Apple Vision Pro, CollabXR offers cross-platform functionality that enables immersive, co-located Mixed Reality (MR) classrooms regardless of device. Over several years of classroom use—particularly in astronomy courses exploring supernova remnant structures—CollabXR has demonstrated increased student engagement and deeper learning. Students pose more sophisticated questions and grasp complex material more quickly, indicating a significant improvement in learning outcomes.

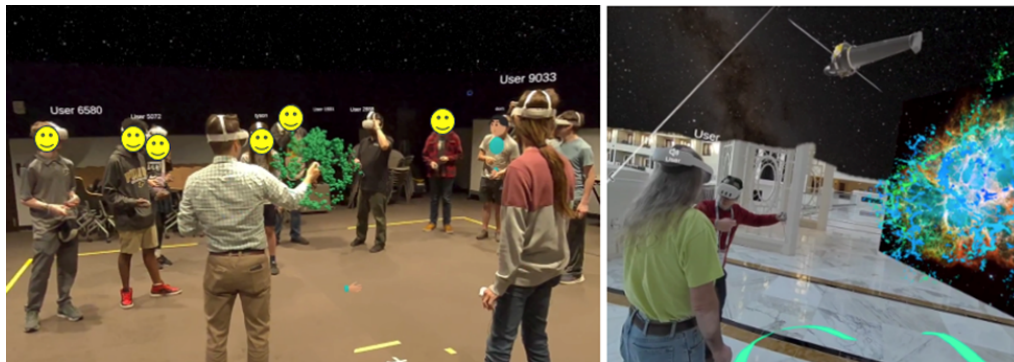
Visualization of complex three-dimensional structures presents a challenge when flattened to two-dimensional illustrations or individual volume slices. In the communication of spatial information—such as shape and perspective—gestures play a key role in supporting verbal explanations. Alibali [3] highlights how gestures complement speech and contribute to comprehension, particularly when conveying spatial concepts. However, social VR platforms often rely on virtual avatars to represent participants, which can limit the fidelity of nonverbal cues. Dissemination and scaling of immersive technologies present challenges across both consumer



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**Figure 2:** Example domain-specific radar meteorology spatial data visualization (left) with cross-sectional representations of internal structures as from cutting planes (middle), and final interactive volumetric representation in a shared MR classroom at Purdue University EAPS. [4]



**Figure 3:** Left: PI-Milisavljevic and Co-PI Takahashi demonstrate CollabXR at Purdue for a local high school. Right: Milisavljevic performs impromptu showcase at the Maryland Gaylord National Resort Hotel lobby during the 2025 AAS conference, demonstrating both power and mobility of the platform.

available resources to state-of-the-art equipment. Even with the lowered barrier to entry, XR continues to present accessibility and content authoring present challenges for broader adoption. To improve classroom experiences and streamline setup, limitations to class group sizes of about 15 per faculty breakout room and automatic space calibration with shared spatial anchors and fiducial marker calibration with a QR code has further facilitated the creation of classes.

## 2 Applications of CollabXR

In Spring 2025, Purdue's Office of the Vice Provost for Teaching and Learning supported a major expansion of CollabXR, enabling its deployment across multiple disciplines, including physics, aerospace, mechanical engineering, and nursing. Faculty noted that the platform's unique affordances—such as shared spatial presence and real-time interaction—necessitate instructional strategies that differ from traditional methods. Simply placing students and instructors in an MR environment is not sufficient; to fully leverage the medium's potential, deliberate learning design and delivery strategies tailored to MR's distinctive capabilities are essential.

CollabXR's open-source architecture ensures that educators, researchers, and developers can freely access, modify, and extend the platform. Compatibility with commercially available hardware and a lightweight installation process will make the technology broadly accessible, including for institutions with limited resources.

Accessibility features—such as adjustable interaction modes, voice-based navigation, and compatibility with assistive devices—will be integrated to ensure participation for all learners. While rooted in STEM, the generalizable pipeline is immediately applicable to other disciplines that rely on 3D visualization, including digital humanities, archaeology, and cultural preservation. Moreover, the portable nature of this technology and compatible hardware will simplify the mobile capacity for this solution to allow traveling exhibits, conferences, and ad-hoc presentations with only the need for a network connection (including local only if also hosting a local relay server).

## 3 Data and Lessons Learned

As the platform extended its use to additional instructors, several faculty members from various fields (e.g., Physics, Languages) noted that the co-located XR environment offers a unique affordance, potentially identifying another reason for the variety of outcomes in XR deployments: It was observed to be insufficient to simply place the lecturer and students in the environment together if the instructional method did not take this affordance into account. Instructional methods in XR can and should differ from traditional approaches, often embracing more experiential and interactive engagement. As XR introduces a powerful new toolkit for education, it is essential to explore how learning design, learning theories, and assessment can be effectively integrated into this emerging medium. Both the instructional strategies that leverage XR and the technical



**Figure 4: Dr. Jun Chen in Mechanical Engineering highlighting vorticity in CFD model of airfoil design**

innovations required to support them must be expanded, refined, and rigorously evaluated to ensure meaningful improvements in teaching and learning.

Feedback from over 300 students who used CollabXR over the fall 2026 semester within veterinary sciences, nursing, physics and astronomy revealed substantial improvement in reducing motion sickness down to less than 5% of participants over 40-minute lectures. This is compared to the 30-35% drop out rates experienced during the VR version in 2020, where users reported nausea and discomfort within the first 30 minutes. Students' interest in the medium showed a high level of engagement both from the survey results, open ended feedback, and the data/faculty interaction within the experiences. Additional feedback from nursing and veterinary students included a desire to further link 3D anatomy with the practical and theoretical materials to better connect the learning materials. As these were using consumer devices additional best practices in sanitizing, device management, and group coordination presented challenges or required additional time.

## 4 Conclusions

The mission of CollabXR is to provide support to the visualization requirements of university faculty for their student instruction. Having advanced the understanding of complex interactions between STEM students' individual differences and immersive technology to inform device and design decisions, these discoveries will cascade through the Envision Center XR Lab's ongoing offering to provide XR devices and immersive experience development to faculty partners seeking to create new educational opportunities for students by guiding immersive practices to maximize the benefit to learners while minimizing risks of engaging with the visualization tool.

Successful implementation of a visualization tool into an educational environment requires understanding how different learners may interact with the product. As a highly immersive solution may carry additional risks to outcomes, which may disproportionately affect those whom the visualization tool is presumably intended to help the most, it is warranted to examine whether the more immersive experience does indeed communicate spatial information more effectively than the less immersive experience. This project addresses the gap by examining the difference in spatial communication potential between the more immersive headset and less

immersive tabletop visualizations, conveying spatial structure data critical to conceptual understanding for a variety of STEM, arts and humanities domains.

The focus on the spatial communication potential is driven by the expectation that, to produce the best educational outcome, the optimal presentation medium will provide students with the advantageous benefit to perceptual understanding while simultaneously imposing the fewest factors that may inhibit comprehension of the material. As the barrier to entry to obtain these technologies continues to decrease, this work will point toward effective solutions that will be accessible to more students as the interest in leveraging XR in schools and other educational programs at various levels increases. To meet this need, the CollabXR platform is aiming towards publication and open-source distribution to create an accessible visualization and learning platform for all.

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